AgentWorkShop — End-to-End Execution: Orchestration → Agent-Internal Workflow → Industry Integration

Initiation
objective → plan

GATE ① plan
approval
locks version

Route Loop
no-LLM dispatch

Acceptance
L0 → L1 → L2

GATE ③
completion
owner signs

completed
never automatic

zoom ↓

↑ results → three-level acceptance

Zoom-in — inside one Channel (agent-internal workflow)

what happens when a dispatched work item arrives: from root task to completion callback

1 · Root task lands
project root → channel FIFO queue
brief · budget · handoff inputs

2 · Lead claims & plans
LLM lead via mailbox · agent-token
reads handoff artifacts

3 · Dispatch workers
dispatch_task on active root
workers = isolated sub-agents

4 · Governed execution
tools: registry / policies (all·selected)
exec mode: observe / manual / auto

5 · HITL on risky actions
approval card → owner / reviewer
observe mode: log only

6 · Aggregate & complete
lead merges worker artifacts
complete_task → COMPLETED

progress: self-report + derived

workers: no cross-channel

lead idle → D1 self-heal

read: DAQ telemetry ↑

act: governed writes (HITL) ↓

Industry integration — how execution touches the physical line

Acquisition path (read)
sensors → DAQ drivers
modbus · opcua · mqtt · http
→ TSDB: batch-tag · retention
serial = plugin · S7 = reject stub

Control path (write)
recipe-bound write intent
throat: range ∩ step ∩ 60s ∩ window
drivers → PLC / actuator → readback
ACK: verified / transport / unverified

MES & twin-first
MES-Rest bridge (write side)
twin mirror → six-step cutover
swap = driverConfig only
HITL card in UI (/dcw/line)

orchestration flow · agent execution · human gate · industry data / control

blue dashed = orchestration & callbacks · orange dashed = human approval · grey = industry data/control